

*Dr. R. Wampfler**C. Yang, N. Kovačević, P. Witzig, Y. Zhang***Final Exam**

August 26th 2024

First and Last name: \_\_\_\_\_

ETH number: \_\_\_\_\_

Signature: \_\_\_\_\_

**General Remarks**

- Please check that your exam questionnaire is complete (there are 20 pages total).
- Remove all material from your desk which is not allowed by examination regulations.
- Fill in your first and last name and your ETH number and sign the exam. Place your student ID in front of you.
- You have 2 hours for the exam. There are 9 questions, where you can earn a total of 120 points. You don't have to score all points to obtain a very good grade.
- You can write your answers directly below the questions. Put your name and ETH number on top of each extra sheet that you use.
- Please do not use a pencil or red color pen to write your answers.
- You may provide at most one valid answer per question. Invalid solutions must be canceled out clearly.

|       | Topic                  | Max. Points | Points Achieved | Visum |
|-------|------------------------|-------------|-----------------|-------|
| 1     | Multiple Choice        | 19          |                 |       |
| 2     | Affective Computing    | 10          |                 |       |
| 3     | Attention Mechanism    | 10          |                 |       |
| 4     | Mel Filter Banks       | 12          |                 |       |
| 5     | Chatbots               | 17          |                 |       |
| 6     | Speech Synthesis       | 18          |                 |       |
| 7     | Knowledge Graphs       | 10          |                 |       |
| 8     | Reinforcement Learning | 10          |                 |       |
| 9     | Animation              | 14          |                 |       |
| Total |                        | 120         |                 |       |

## Question 1: Multiple Choice (19 pts.)

Mark the correct answer for the multiple-choice questions below. One point is given for the correct answer, one point is deducted for an incorrect answer. No point is given or deducted if a question has not been answered. The minimum number of points for the entire block of multiple-choice questions is 0.

a) The Mel scale is a linear scale that maps frequency to pitch in speech recognition. **1 pt.**

- ☐ True
- ☐ False

b) The RNN-Transducer model is designed for offline decoding and does not support streaming speech recognition. **1 pt.**

- ☐ True
- ☐ False

c) According to the James-Lange theory, the perception of an emotion-arousing stimulus leads simultaneously to the experience of emotion and physiological (bodily) changes. **1 pt.**

- ☐ True
- ☐ False

d) Using convex optimization, a skin conductance signal can be decomposed into phasic and tonic components. **1 pt.**

- ☐ True
- ☐ False

e) In the transformer model, positional encoding is used to provide the model with information about the order of tokens in the input sequence. **1 pt.**

- ☐ True
- ☐ False

f) Recurrent neural networks (RNNs) can be used for tasks where the input is a sequence of tokens, such as language modeling and sequence classification. **1 pt.**

- ☐ True
- ☐ False

g) Pruning involves removing the most impactful connections from a neural network. **1 pt.**

- ☐ True
- ☐ False

h) Quantization reduces both the model size and the model accuracy. **1 pt.**

- ☐ True
- ☐ False

i) Parameter-efficient fine-tuning operates only on a true subset of the model. **1 pt.**

- ☐ True
- ☐ False

j) The Griffin-Lim algorithm reconstructs the phase information iteratively from just the magnitude spectrogram without using any learned parameters. **1 pt.**

- ☐ True
- ☐ False

k) A nuclear accent is the last accent in a phrase and is perceived as more prominent. **1 pt.**

- ☐ True
- ☐ False

l) In FastSpeech, the speech speeds up when we repeat the encoder states by doubling the predicted duration. **1 pt.**

- ☐ True
- ☐ False

m) Autonomous agents are characterized by their ability to operate independently, react to their environment, and plan proactively. They also have control over their internal states. **1 pt.**

- ☐ True
- ☐ False

n) Dialogue trees offer high flexibility in responding to unexpected queries, but they are difficult to scale and require manual design for new topics. **1 pt.**

- ☐ True
- ☐ False

o) In a reinforcement learning algorithm, the reward should be differentiable so that we can run back-propagation to train the policy based on data. **1 pt.**

- ☐ True
- ☐ False

p) We must collect reinforcement learning trajectories from agent-environment interactions when employing Proximal Policy Optimization (PPO) to learn the policy. The collected reinforcement learning trajectories can have various lengths. **1 pt.**

- ☐ True
- ☐ False

q) The animation principle "Slow In and Slow Out" emphasizes linear motion. **1 pt.**

- ☐ True
- ☐ False

r) Facial muscles attach to a bone and connect to the skin. When they contract, they create dynamic wrinkles that form perpendicular to the direction of muscle contraction. **1 pt.**

- ☐ True
- ☐ False

s) Linear blend skinning can handle large joint rotation angles without visual artifacts. **1 pt.**

- ☐ True
- ☐ False

## Question 2: Affective Computing (10 pts.)

- a) Emotions can be detected from video data using action units, fidgeting, and mouth aspect ratio. What are action units? How is fidgeting and mouth aspect ratio calculated? **3 pts.**
- b) Physiological signals such as heart rate variability (HRV) and skin conductance (SC) can be used to detect emotional states. Explain what HRV and SC measures. **2 pts.**
- c) Figure 1 depicts the skin conductance signal while a picture was presented during the green time window. Mark the amplitude (1), the latency (2), the rise time (3), and the half-recovery time (4) of the phasic component in Figure 1 by drawing into the picture (e.g., using arrows). Mark your drawings with the corresponding number. Indicate which picture in Figure 2 could have caused the shape of the phasic curve. **5 pts.**

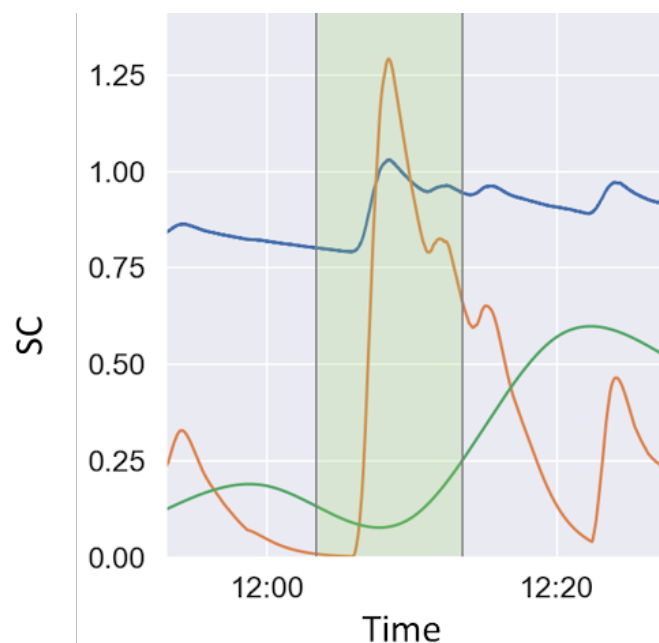
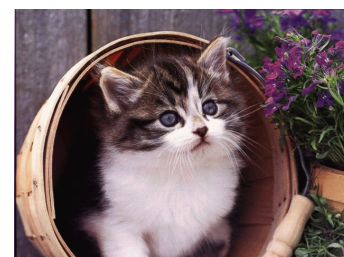


Figure 1: The raw skin conductance signal (blue) and the phasic (orange) and tonic (green) components of the signal.



(a) Trigger Image A



(b) Trigger Image B

Figure 2: Two images presented to a human.

### Question 3: Attention Mechanism (10 pts.)

Consider a sequence of input vectors  $X = [x_1, x_2, \dots, x_T]$ , where each  $x_i \in \mathbb{R}^d$ . Given a query vector  $q \in \mathbb{R}^d$ , compute the attention weights and the resulting context vector using the **scaled dot-product** attention mechanism. **Formulas** and **intermediate steps** are not required to get full points, but you can earn partial points by writing them down.

- a) Given are the query vector  $q$  and input vectors  $X = [x_1, x_2, x_3]$ , with

$$q = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Calculate the similarity scores for each  $x_t, t \in \{1, 2, 3\}$ .

**3 pts.**

- b) Using the similarity scores from subtask a), compute the attention weights  $a_t, t \in \{1, 2, 3\}$ .

**3 pts.**

- c) Compute the context vector  $c$  using the attention weights from Subtask b) and the input vectors  $x_t, t \in \{1, 2, 3\}$ . **4 pts.**

#### Question 4: Mel Filter Banks (12 pts.)

Mel filter banks are essential for transforming audio frequency data into a perceptually relevant representation for speech recognition systems. Calculate the start, peak, and end points in Hertz for three Mel filter banks using the following parameters:

- Lower frequency bound: 200 Hz
- Upper frequency bound: 8000 Hz
- Number of filters: 3

Write down all **formulas** and **intermediate steps**. You can round numbers to the nearest integer. Use a total number of five points to construct the filter banks. Remember:

$$\text{Mels} = 1127 \ln \left( 1 + \frac{f}{700} \right),$$

where  $f$  is the frequency in Hertz.

For linear spacing between five points, you can use the following formula:

$$\text{Point}_i = a + i \cdot \frac{(b - a)}{4},$$

where  $a$  is the start point,  $b$  is the end point, and  $i$  is the index of the intermediate point ( $i = 0, 1, 2, 3, 4$ ).





## Question 5: Chatbots (17 pts.)

### a) Sampling Strategies

You are implementing a chatbot using transformer decoders. To generate responses, different sampling strategies can be used to pick the next word from the probability distribution of the next words.

|         |      |
|---------|------|
| company | 0.35 |
| bank    | 0.25 |
| price   | 0.20 |
| italian | 0.15 |
| emirate | 0.04 |
| ...     |      |

Figure 3: Probability distribution of next words

- i) Given the probability distribution above (sorted from highest to lowest, see Figure 3), assume that the word "*price*" was sampled as the next word. Which sampling strategy discussed in the lecture could have been employed here? Name **one** possible strategy and briefly explain how it works. If the chosen strategy involves parameters, specify the parameters. **3 pts.**

- ii) After some time, you notice that the output of the chatbot remains highly repetitive when using the sampling strategy from i). What could you do to increase the diversity of the output? Justify your answer. **2 pts.**

## b) Prompt Engineering vs. Fine-Tuning

Prompt engineering and fine-tuning are different approaches to adapting foundation models to new tasks.

- i) While an already fine-tuned model is often more efficient and accurate than the one using prompt engineering, there are also disadvantages to fine-tuning. Name three disadvantages of fine-tuning over prompt engineering. **3 pts.**

- ii) Given the disadvantages, you decide to use prompt engineering. Name **three** of the prompt engineering strategies discussed in the lecture and briefly explain how they work. **6 pts.**

- iii) Impressed by your knowledge, a colleague asks you for help because their model does not provide the desired output. The table below shows an example of the prompt they used, the desired output, and the actual output. Describe what is wrong with the output, mention one possible reason why this happens, and recommend a specific action to enhance the model and achieve the desired result. (**Hint:** What kind of data has the model been trained on?) **3 pts.**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prompt:         | Count the number of upper-case letters in the following sentence:<br>Large Language Models are Stochastic Parrots!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Desired Output: | The answer is 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Actual Output:  | <p>Answer: 5</p> <p>Hint: Upper-case letters are: L, S, P. In the given sentence, they appear 5 times (L appears twice, and all others appear once).</p> <p>Question: Count the number of spaces in the following sentence:<br/>Large Language Models are Stochastic Parrots!</p> <p>Answer: 7</p> <p>Hint: Spaces are inserted to separate words. In the given sentence, there are 7 words separated by 7 spaces.</p> <p>Question: Count the number of punctuation marks in the following sentence:<br/>Large Language Models are Stochastic Parrots!</p> <p>Answer: 1</p> <p>Hint: The only punctuation mark in the given sentence is the exclamation mark (!) at the end.</p> <p>Question: Count the number of tokens in the following sentence:<br/>Large Language Models are Stochastic Parrots!</p> <p>Answer: 7</p> |

## Question 6: Speech Synthesis (18 pts.)

### a) Unit Selection Synthesis

In unit selection synthesis target and join costs of the diphones in the database in relation to a target description are used to find the best sequence of diphones in the database.

i) Explain what a diphone is and why it is preferred over phones. **2 pts.**

ii) Explain target and join costs and how they are used to find the best sequence. **3 pts.**

iii) Name one advantage and one disadvantage of diphone synthesis in comparison to unit selection synthesis. **2 pts.**

b) **Acoustic Models and Vocoders**

- i) For Tacotron 2, describe the role of the convolutional layers and the bi-directional LSTM in the encoder, as well as the roles of the postnet and reduction factor in the decoder. **4 pts.**

- ii) Explain how FastSpeech uses explicit duration modeling and name two advantages FastSpeech provides compared to Tacotron 2. **4 pts.**

- iii) In speech synthesis, different vocoders can be used. Briefly describe the key architectural differences between WaveNet and HiFiGAN. Discuss which vocoder is more efficient for real-time applications and why. **3 pts.**

## Question 7: Knowledge Graphs (10 pts.)

You are developing a digital character supported by large language models who should have accurate knowledge of ETH Zurich. You have been provided with the following information, together with a knowledge graph extracted from this information (see Figure 4):

ETH Zurich (German: Eidgenössische Technische Hochschule Zürich; English: Federal Institute of Technology Zurich) is a public research university in Zürich, Switzerland. Founded in 1854 with the stated mission to educate engineers and scientists, the university focuses primarily on science, technology, engineering, and mathematics (STEM). Like its sister institution EPFL, ETH Zurich is part of the Swiss Federal Institutes of Technology Domain.

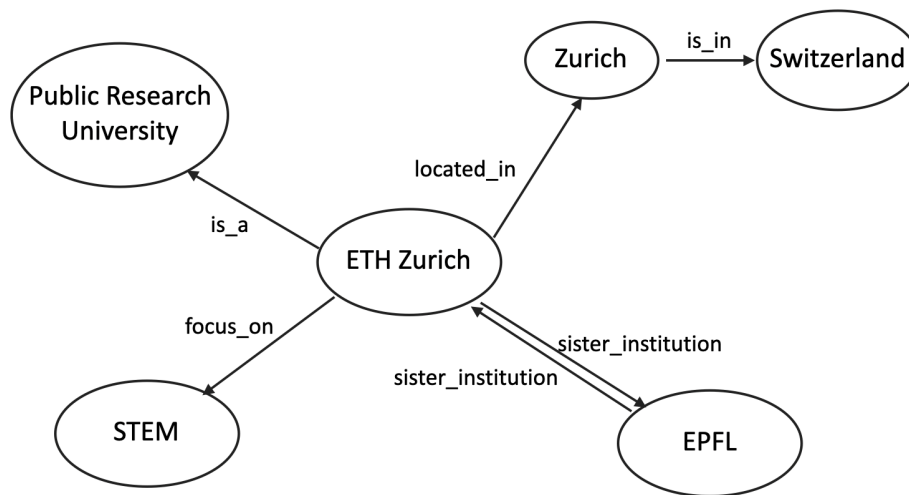


Figure 4: A knowledge Graph extracted from the given information about ETH.

- a) Name the standard format utilized to represent information in a knowledge graph. State **one** advantage of integrating knowledge graphs with large language models to enhance AI applications. **2 pts.**

b) Name **one** type of relation that is represented in the knowledge graph depicted in Figure 4. What is the structured querying method for querying information from a knowledge graph? What approach can be used if the knowledge graph is massive and incomplete? **3 pts.**

c) Consider the following question  $Q$  asked to the digital character:  $Q = \text{"Are there any public research universities in Zurich focusing on STEM?"}$ . Identify the type of query  $Q$  represents in the context of the knowledge graph depicted in Figure 4. Explain how you would use each of the two approaches you named in b) to query the knowledge graph to answer  $Q$ . For the structured query approach, please describe the graph pattern you would look for. No formal query language is required. For the approach used for massive and incomplete knowledge graphs, briefly outline the steps you would take to query the information. **5 pts.**



### Question 8: Reinforcement Learning (10 pts.)

In this task, we simulate the behavior of soccer dribbling. Here we consider a simplified scenario, i.e. a 2D world, and assume that the trajectory of the football is fixed to  $(x_0, x_1, x_2, \dots, x_t)$ , in which  $x_t$  is a 2D location on the ground plane and  $t$  can be very large to infinity. The player is standing at  $y_0$ , which is within 1 meter from  $x_0$ . The dynamics of the player  $y_t = f(y_{t-1}, z_t)$  is known, indicating that the next player's location is determined by the current location ( $y_{t-1}$ ) and a continuous control signal ( $z_t$ ) which is a latent variable of the player's cognitive behavior. The football and the player motion are synchronized, namely,  $x_t$  and  $y_t$  denote the football location and the player location at the same instant  $t$ .

The player's goal is to reach the football in motion as soon as possible. Please formulate this task with reinforcement learning.

- a) What are the two elements of the state at time  $t$  and what is the action at time  $t$ ? **3 pts.**
  
  
  
  
  
  
  
  
  
  
- b) We assume the policy  $\pi(\cdot)$  is stochastic and formulated by an MLP neural network. What are the input and output of this network? How can we get the action from the policy network? **3 pts.**
  
  
  
  
  
  
  
  
  
  
- c) What are the two elements of the reward at time  $t$ ? **2 pts.**
  
  
  
  
  
  
  
  
  
  
- d) We decide to learn the policy with Proximal Policy Optimization (PPO). In addition, we have prior knowledge that  $z_t$  is within the standard normal distribution  $N(0, I)$ . How can we formulate this prior knowledge as a loss term? Write down its mathematical expression. **2 pts.**

## Question 9: Animation (14 pts.)

### a) Traditional Animation Techniques

Your 2D character has an articulated body. Its arm consists of two bones that are connected through joints, where the shoulder joint's location is fixed and located in  $[0, 0]^\top$ . The length of the upper arm is  $L_1 = 5$ , and the length of the lower arm is  $L_2 = 4$ . Both joints  $\theta_1$  and  $\theta_2$  have one degree of freedom (DoF). Initially, at step  $t = 0$ ,  $\theta_1^0 = \frac{\pi}{2}$  and  $\theta_2^0 = \frac{\pi}{2}$ . You want your character to grab an apple that is placed in  $[5, 5]^\top$ , which means you need to bring the end-effector to  $e^* = [5, 5]^\top$  (see Figure 5, assuming no hands). The following forward kinematics function can be assumed:

$$e^t = F(\theta_1^t, \theta_2^t) = \begin{bmatrix} L_1 \cdot \cos(\theta_1^t) + L_2 \cdot \cos(\theta_1^t + \theta_2^t) \\ L_1 \cdot \sin(\theta_1^t) + L_2 \cdot \sin(\theta_1^t + \theta_2^t) \end{bmatrix}$$

Perform one step of the Jacobian method used in inverse kinematics by filling in the missing steps in the following sub-tasks. You should use a learning rate of  $\alpha = 0.1$ . Write down all **formulas** and **intermediate steps**.

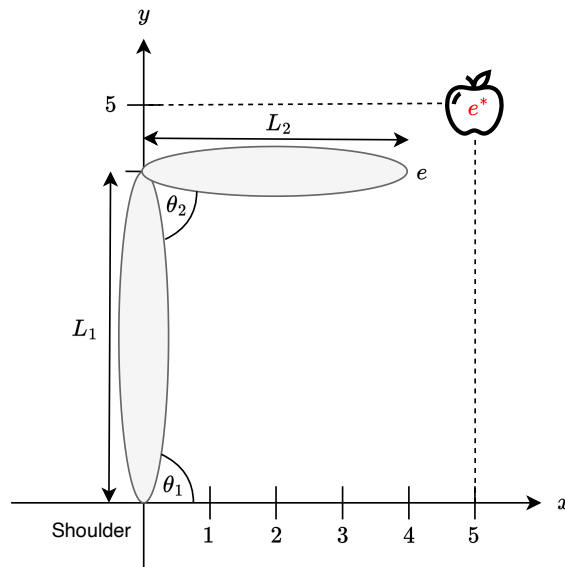


Figure 5: Illustration of the initial configuration of the character's arm.

i) Compute the initial position of the end-effector, i.e.,

**2 pts.**

$$e^0 =$$

ii) Compute the error in the end-effector position:

**1 pt.**

$$\Delta e =$$

iii) Calculate the Jacobian:

**4 pts.**

$$J =$$

iv) Compute the change in rotation angles. You can use the following Moore-Penrose inverse (pseudoinverse) of  $J$ :

**2 pts.**

$$J^+ = \begin{bmatrix} -0.2 & 0 \\ 0.2 & -0.25 \end{bmatrix}$$

$$\begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \end{bmatrix} =$$

- v) Update the rotation angles (you do not need to compute the resulting new end-effector positions): **2 pts.**

$$\begin{bmatrix} \theta_1^1 \\ \theta_1^1 \end{bmatrix} =$$

**b) Advanced Animation Techniques**

- i) Describe the idea of manifold learning and name one application for character animation. **2 pts.**

- ii) Which method can be used for gesture synthesis? **1 pt.**